

Why don't AI/data centers cool their computers with salt water?

Posted by u/fuckmeup-scotty • 0 points • 6 comments

I see it all the time - new AI data centers basically running a towns water supply dry, using SO much fresh water that it's causing severe shortages and worsening droughts. So why do they HAVE to use fresh water? Is it a choice? There's an unbelievable amount of salt water out there, so why not just build close to the ocean and pump that water? I know it's more corrosive, but we have so many things that stay in the ocean, so we've definitely figured out some materials that are more resistant and won't need to be replaced as frequently.

Does it not cool things as quickly/efficiently? Why not? This seems like it could be a great solution, but I'm not seeing anyone doing it

Comments

u/ResponsibilityNo8309 • 2 points • 2025-10-07 14:58 UTC

Salt water is corrosive and will cause blockages in cooling systems if not regularly cleaned. It would also require that all data centres would have to be right on costal land otherwise they would use significantly more energy pumping the salt water than just using water. Also it would make data centers more vulnerable to storms.

u/Notmiefault • 1 points • 2025-10-07 14:56 UTC

Salt is extremely corrossive and damaging to electronics.

u/Quixlequaxle • 6 points • 2025-10-07 14:30 UTC

Basically three issues:

- 1) Salt water is quite corrosive, so needing pumps, piping, heat exchangers, etc to be resistant to that is quite expensive
- 2) Increased risk of flooding if you're building data centers on the coast
- 3) Data centers use a lot of electricity, and electricity is much less expensive in the small-town areas where they are building data centers than it is in coastal areas

u/sexrockandroll • 4 points • 2025-10-07 14:29 UTC

Salt is super corrosive.

u/PresentDirection41 • 6 points • 2025-10-07 14:27 UTC

> I see it all the time - new AI data centers basically running a towns water supply dry

You absolutely don't see this "all the time." This has happened like once at most.

u/bangbangracer • 15 points • 2025-10-07 14:24 UTC

Because salt.

It builds up in the lines over time and increases corrosion.